INDEPENDENT RESEARCH TECHNICAL NOTE

Correct outputs, contaminated search

A state-ownership contract for executable world models

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Can a planner trust a model call that changes the parent state of its hypothetical branches?

I identified a state-ownership failure in released executable world models: a call can return the same state values as a full-copy reference while modifying nested objects owned by its caller. Sibling branches then start from different parents, making search sensitive to rollout order.

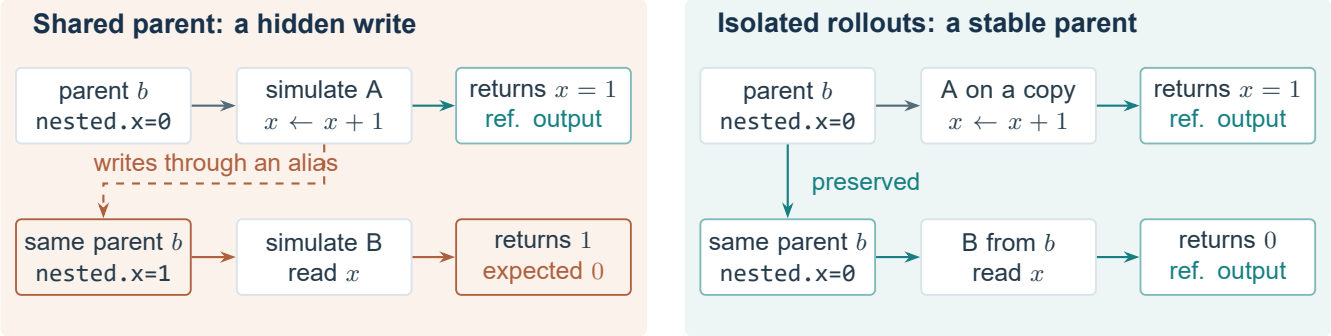


Figure 1. Illustrative deterministic program. Action A's return matches the full-copy reference in both panels; only the isolated parent remains reusable by B.

01 The mechanism I implemented: WCTR

I implemented **Witness-Compiled Transactional Rollouts** to learn mutation paths from executions, selectively fork their alias ancestors, and mark new allocations private. Unexpected writes are journaled; output grafting preserves the returned successor before parent rollback.

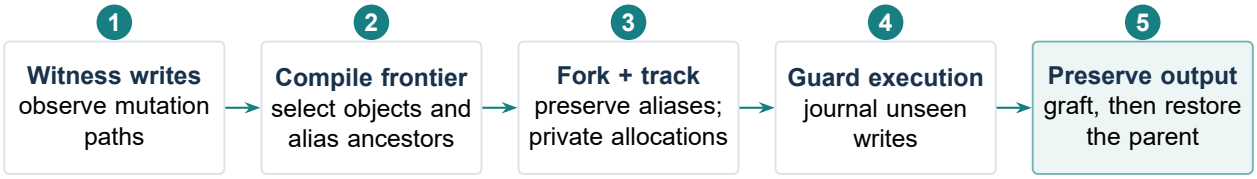


Figure 2. Witnesses guide selective copying; runtime guards handle unseen writes. Alias preservation and allocation tracking are part of the implementation.

Operational target: an unchanged parent and full-copy-equivalent values

$$b_{\text{after}} \equiv b_{\text{before}}, \qquad b'_a \equiv F(\text{deepcopy}(b), a)\,.$$

Equivalence is with the **same program**, not ground-truth prediction. Checks cover instrumented dict, list, set and tuple operations; they establish neither arbitrary-Python isolation nor theoretical novelty.

59 / 3,942
Discovery in BabyAI
input mutations in recorded predictions

0 / 33,692
Checked WCTR transitions
parent/output mismatches in the frozen live study

Finite workload checks; other protected controls also had zero parent/output mismatches.

SELECTED EVIDENCE / PROPOSED RESEARCH DIRECTION

From branch isolation to decision trust

I used controlled comparisons to separate an implementation's branch behavior, bounded planning performance, and actual interactive task completion.

02 A controlled planning result, with its boundary

Full Copy	WCTR	
47.1%	64.6%	+17.5 pp
WebShop program-goal completion		95% paired interval: +10.6 to +24.8 pp

Exact setting. 80 generated structured-state goals, three timing repeats, shared BFS (depth at most six; at most 4,096 expansions), and a **soft 10 ms planning deadline excluding setup and calibration**. Completion was checked by independent full-copy execution of the released program. The interval resamples source items in the frozen paired analysis.

This is **not an improvement in native WebShop success**. Controlled outcomes were mixed: Wordle declined from 54.2% to 45.0% (**-9.2 pp**) under the same protocol.

Native-task takeaway. Full Copy, Top-Field, Journal Only and WCTR all completed **6/12 WebShop goals** and **43/100 Wordle seed tasks**. There was no incremental live-success benefit. With lifetime setup, verification and profiling charged, simpler protected controls were cheaper overall on this host; no general efficiency or superiority claim follows.

Native scope: small 1k-product WebShop catalog; released 40-word stochastic Wordle. The complete frozen archive remains authoritative.

03 A small next experiment

Which simulator errors can change the action a planner chooses?

Proposed future work; not executed.

1. **Fix the unit.** Snapshot one belief/model state. Fix three candidate actions, a leaf-value rule and canonical tie-breaking.
2. **Vary only root order.** Reset before each of six root permutations; keep child order fixed. Evaluate depth two with 12 expansions (three roots plus nine children), without early stopping; record actual CPU/setup.
3. **Pair the randomness.** Compare unprotected execution, Full Copy and WCTR. Begin with a deterministic case. For controllable stochastic programs, replay five predeclared random tapes keyed to action history across all methods and orders.
4. **Measure the decision effect.** Record pairwise ranking inversions, chosen-action flips versus Full Copy, and parent-state fingerprints. Within-tape order effects with parent mutation and a stable reference implicate contamination.
5. **Separate stochasticity.** Use Full Copy's across-tape variation as the stochastic control. If its ranking changes with order under the same tape, flag uncontrolled randomness or side effects rather than attribute the change to state contamination.

RAP and LLM Reasoners [1,2] motivate studying the model/search interface. I would like to connect execution checks to decision relevance [3], so finite verification effort targets simulator errors that could change an action.

Selected connections [1] Hao et al. RAP. EMNLP 2023. [2] Hao et al. LLM Reasoners. 2024.

[3] Farahmand, Barreto & Nikovski. Value-Aware Loss Function for Model-based Reinforcement Learning. AISTATS 2017.